Solving the static Einstein equations with physics-informed neural networks

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A classical existence theory guarantees that the stationary vacuum Einstein equations, posed in the region outside a ball, have a unique solution for boundary data close enough to flat space, depending continuously on the data. We present a physics-informed neural network (PINN) solver for this boundary-value problem in the static case. The unknowns are a Riemannian 3-metric h and the squared norm λ of the static Killing vector, in harmonic gauge; the data are λ and the induced metric on an inner sphere, and asymptotic flatness is imposed by Robin conditions on an outer sphere. The connection is the Christoffel symbol of the network metric. We test the solver on Schwarzschild in harmonic coordinates and on two-rod Weyl solutions, on the symmetry axis and rotated so that no symmetry is left. For quadrupolar data, which have no closed-form solution, we compare with an independent spectral solution of the same problem. The harmonic gauge and the inner data leave the asymptotic chart free to translate, and this translation dominates the error until it is fixed by Cederbaum's pseudo-Newtonian centre of mass, a conserved flux of the static equations; with it, the solution agrees with the spectral one to five to six orders below the size of the perturbation. Finally, we read the Geroch–Hansen mass moments of the computed solutions off the symmetry axis, without constructing Weyl coordinates; they reproduce the exact moments of the Weyl solutions and the reference moments of the quadrupolar one.

I. INTRODUCTION

For many years there has been a mathematical theory of the stationary Einstein equations in an asymptotic region of spacetime, namely the region outside a ball [1]. It states that boundary data close enough to those of flat spacetime, together with the asymptotic conditions, produce a unique solution that depends continuously on the data. Very few attempts have been made to explore this theory numerically [2]. Here we show that this is possible, and that it opens a new window on these spacetimes.

The method also gives a simple way to explore other systems of possible physical importance: systems in which one solves for a Riemannian metric together with other fields, through equations that contain its Ricci tensor. Examples are the construction of minimal-energy data sets [3] and the equations that give higher-order Newtonian-limit expansions [4]. [TODO: more examples]

We restrict this work to the static Einstein equations. The stationary case is no harder from the point of view of the approximation method, and is left for a continuation of this work.

Physics-informed neural networks [5] represent the unknown fields by a neural network and minimise the squared residual of the equations and boundary conditions over sampled points. They need no mesh, handle the non-compact exterior through a logarithmic radial variable, and treat a configuration with no symmetry exactly as they treat a symmetric one. A small residual does not by itself mean a small error, so every result below is graded against an exact solution or an independent reference.

Section II states the equations, the gauge, the boundary and asymptotic conditions, and the residual gauge freedom. Section III describes the solver. Section IV tests it on exact solutions, Sec. V solves the quadrupolar problem, and Sec. VI extracts the multipole moments. Appendix A lists what is needed to repeat every computation.

II. THE STATIC EINSTEIN EQUATIONS

A. Variables and field equations

We write a static vacuum spacetime as

$$g = -\lambda\,\mathrm{d}t^2 + \lambda^{-1}h, \tag{1}$$

where h is a Riemannian metric on the spatial slice and $\lambda = -\xi\cdot\xi$ is the squared norm of the static Killing vector $\xi = \partial_t$; if $g = -N^2\mathrm{d}t^2 + \gamma$, then $\lambda = N^2$ and $h = N^2\gamma$. The vacuum equations become

$$R_{ab}(h) = \frac{1}{2\lambda^2}\,\nabla_a\lambda\,\nabla_b\lambda, \tag{2}$$

$$h^{ab}\nabla_a\nabla_b\lambda = \frac{1}{\lambda}\,h^{ab}\nabla_a\lambda\nabla_b\lambda. \tag{3}$$

With $U = \frac{1}{2}\ln\lambda$ these read $\mathrm{Ric}(h) = 2\,\mathrm{d}U\otimes\mathrm{d}U$ and $\Delta_h U = 0$.

In some cases we shall solve for the first system of equations, in some cases for the second equivalent system, depending on the problem. The difference being that U is closer to the Newtonian potential, and λ is the squared norm of the Killing vector. Thus, for instance the first multipole moments of U are closer to the Newtonian ones. There are differences, ofcourse, and in fact one finds the $\frac{1}{\rho^2}$ term in λ but not in U.

B. Gauge

We impose harmonic coordinates on h,

$$\Gamma^i \equiv h^{jk}\Gamma^i{}_{jk} = S^i, \tag{4}$$

with $S^i = 0$; equivalently $\Delta_h x^i = 0$ for the Cartesian-like coordinates $x^i = \rho\, n^i$. For the Weyl tests we instead use a source built from the candidate's own metric,

$$S^i = (h_{\rho\rho} - 1)\, h^{ij}\partial_j \ln \rho_n, \qquad \rho_n = \big|x - (x\cdot n)\, n\big|, \tag{5}$$

with n the symmetry axis, which Weyl's canonical form $h = e^{2k}(\mathrm{d}\rho^2 + \mathrm{d}z^2) + \rho^2\mathrm{d}\phi^2$ satisfies exactly.

C. Test solutions

There are known solutions for this system, we shall use some of them to test our solver.

1. *The exact spherical family*

In the harmonic chart the spherically symmetric solutions are

$$h_{ij} = \Big(1 - \frac{R_0^2}{\rho^2}\Big)\delta_{ij} + \frac{R_0^2}{\rho^2}\, n_i n_j, \qquad \lambda = k\,\frac{\rho - R_0}{\rho + R_0}, \tag{6}$$

i.e. $h = \mathrm{d}\rho^2 + (\rho^2 - R_0^2)\,\mathrm{d}\Omega^2$. With $k = 1$ this is Schwarzschild in harmonic coordinates: $M = R_0$, areal radius $r = \rho + R_0$, $\lambda = 1 - 2M/r$. The harmonic condition still allows the radial maps $\rho \to c_1\rho + c_2 F_2(\rho)$, $F_2 = 2R_0 + \rho\ln\frac{\rho - R_0}{\rho + R_0}$, fixed by the round inner metric and $h \to \delta$.

2. *Weyl two-rod solutions*

Static axisymmetric vacuum metrics can be written in Weyl's canonical form [6],

$$g = -e^{2U}\mathrm{d}t^2 + e^{-2U}\big[e^{2k}(\mathrm{d}\rho_c^2 + \mathrm{d}z^2) + \rho_c^2\mathrm{d}\phi^2\big], \tag{7}$$

where ρ_c is the cylindrical radius, U is an axisymmetric solution of the flat Laplace equation, and k follows from U by a line integral, $\partial_{\rho_c}k = \rho_c\,[(\partial_{\rho_c}U)^2 - (\partial_z U)^2]$, $\partial_z k = 2\rho_c\,\partial_{\rho_c}U\,\partial_z U$, with $k \to 0$ at infinity. In our variables $\lambda = e^{2U}$ and $h = e^{2k}(\mathrm{d}\rho_c^2 + \mathrm{d}z^2) + \rho_c^2\mathrm{d}\phi^2$. A rod of length L on the axis, $U_{\rm rod} = \frac{1}{2}\ln[(R_a + R_b - L)/(R_a + R_b + L)]$ with R_a, R_b the distances to its ends, is the Schwarzschild solution of mass $L/2$; two rods give two black holes on the axis (held apart by a strut, where $k \neq 0$ between them). We evaluate k by Gauss–Legendre quadrature. The Weyl chart is not harmonic, but it satisfies the gauge condition (4) with the source (5), which is what we impose in these tests. The inner boundary is a coordinate sphere $\rho = \rho_{\rm in}$ large enough to enclose both rods, and the data on it, λ and the induced metric, are those of the exact solution.

To test a configuration with no symmetry we rotate the solution by an angle α about the y axis. The equations are covariant, so $h \to R\,h\,R^T$ and $\lambda \to \lambda$, evaluated at $R^T x$, is again a solution. The inner data are taken from the rotated fields, and the gauge source (5) is taken about the rotated axis $n = R\hat{z}$: the gauge condition is a statement about the chart and rotates with the configuration.

D. Boundary and asymptotic conditions

The domain is the shell $\rho_{\rm in} \le \rho \le \rho_{\rm out}$.

a. Inner sphere. We prescribe λ and the induced metric,

$$\lambda = \lambda_0 + S_1\, n\cdot e + S_2\, P_2(n\cdot e), \qquad PhP = \frac{r_{\rm in}^2}{\rho_{\rm in}^2}\,P, \tag{8}$$

with $P = I - n\otimes n$, $r_{\rm in}$ the areal radius of the sphere in h and e the data axis. No condition is imposed on h_{rr} or h_{rA}. In harmonic gauge these are determined by the solution itself, and prescribing them would leave no solution of the boundary-value problem except when the prescribed values happen to coincide with the solution's own, which are not known in advance. Numerically this shows up as a loss that stalls instead of decreasing: the spectral solution, with $h_{rA} = 0$ imposed, stopped at a residual of 1.8×10^{-5} with a non-decaying $l = 2$ part, and reached 3×10^{-9} once the condition was removed. λ_0 is chosen so that $\lambda \to 1$ on the spherical branch. In the traditional approach to the problem one would look for boundary conditions that would imply the enforcement of the harmonic gauge condition once the equations are solved. In our approach the harmonic gauge condition is imposed directly, as one of the residuals of the loss in the interior (Sec. III), together with the field equations; the boundary conditions then only have to select a unique solution among those of the equations and the gauge condition. This considerably simplifies their choice, since no conditions are needed whose role is to propagate the gauge.

b. Outer sphere. Asymptotic flatness is imposed by Robin conditions of order n,

$$B_n[f] \equiv \prod_{i=0}^{n-1}\big(\rho\,\partial_\rho + p_f + i\big)\,(f - f_\infty) = 0, \tag{9}$$

which annihilate the decaying powers $\rho^{-p_f}, \ldots, \rho^{-p_f-n+1}$. The exponents follow from the equations, $\delta h \sim \rho^{-2}$, $\Gamma \sim \rho^{-3}$, $\lambda - 1 \sim \rho^{-1}$; the exact solution (6) gives local decay rates 2.00, 3.005 and 0.952 at $\rho = 20$. For the scalar we impose the condition on the Geroch–Hansen potential [7, 8]

$$\mathcal{P} = \frac{1 - \lambda^2}{4\lambda} = -\tfrac{1}{2}\sinh 2U, \tag{10}$$

whose monopole $M/\rho + M^3/\rho^3 + \dots$ has no ρ^{-2} term, unlike $\lambda = 1 - 2M/\rho + 2M^2/\rho^2 - \dots$. [The ladder and the averaged pins below: kept or removed after the ablations ab_noladder, ab_nopins, ab_none.] The same conditions are imposed on a few spheres inside $\rho_{\rm out}$ (at 0.7, 0.8, $0.9\,\rho_{\rm out}$): they hold at every radius for the exact solution, and a single sphere can be satisfied by a thin boundary layer. For the quadrupolar problem we add spherical means of order-one Robin combinations at $\rho_{\rm out}$, $\langle(\rho\partial_\rho + 3)(h_{rr}-1)\rangle$, $\langle(\rho\partial_\rho+2)(g_2-1)\rangle$ with $g_2 = \frac{1}{2}(\mathrm{tr}\,h - h_{rr})$, and $\langle(\rho\partial_\rho + 1)\mathcal{P}\rangle$; without them h drifts towards a multiple of the identity. On λ itself such a pin would be violated by the exact solution, by -6.59×10^{-5} at $\rho_{\rm out} = 100$ (a loss floor of 4.3×10^{-7}); on $\mathcal{P}$ the floor is 1.5×10^{-11}. $\lambda_\infty = 1$ is fixed.

E. The residual gauge freedom and the centre of mass

The harmonic condition fixes the chart only up to harmonic diffeomorphisms $x \to x + \xi$, $\Delta_h\xi = 0$. One of them is left free by our data: $\xi = 0$ on the inner sphere and $\xi \to a$ at infinity. It keeps the chart harmonic and the inner data unchanged, and translates the asymptotic chart by a. Its effect on the scalar, $\xi\cdot\nabla\lambda$, vanishes at $\rho_{\rm in}$ and decays like ρ^{-2}, so the outer conditions barely penalise it: for $|a| \approx 0.04$ it changes the dipole of $\mathcal{P}$ at $\rho = 100$ by $\approx 2.5\times10^{-6}$.

It is fixed by the centre of mass. Since U and the coordinates x^i are both h-harmonic, Green's identity makes

$$D^i(S) = \oint_S \left(x^i\,\partial_\nu U - U\,\partial_\nu x^i\right) \mathrm{d}A_h \qquad (11)$$

the same on every closed surface S enclosing the inner sphere, and $D = 4\pi M a$ for a mass M displaced by a. This is the pseudo-Newtonian centre of mass that Cederbaum introduced for static ("geometrostatic") isolated systems [9], written in our variables. It is quasi-local, obeys an addition theorem for several separated sources, coincides with the ADM and constant-mean-curvature centres of mass in suitable asymptotic coordinates, and converges to the Newtonian centre of mass in the Newtonian limit [9, 10]. The asymptotic notions it is compared with go back to Regge–Teitelboim and Beig–Ó Murchadha [11, 12] and to the foliation-based definitions of Huisken–Yau, Metzger, Huang, Corvino–Wu and Eichmair–Metzger [13–18] (see Ref. [19]); Cederbaum's analysis builds on the asymptotics of static fields of Beig, Beig–Simon and Kennefick–Ó Murchadha [20–22]. We impose $d \equiv D/4\pi = 0$ on a set of spheres. This is a gauge choice, available for any data; Sec. V shows that without it the translation mode is the largest error of the computed solution.

III. THE SOLVER

a. Networks. The inputs are $n = x/\rho$ and $t = \ln(\rho/\rho_{\rm in})/\ln(\rho_{\rm out}/\rho_{\rm in}) \in [0,1]$, optionally with $\rho_{\rm in}/\rho$; the body is a tanh multilayer perceptron, the output layer is initialised small so that training starts near flat space. The network outputs h and $\lambda = e^u$, and the connection is the Christoffel symbol of h, computed by automatic differentiation, so the compatibility equation holds identically. (A first-order variant outputs Γ as well and adds compatibility as a residual; on the same Schwarzschild problem it is three orders of magnitude less accurate, Sec. IV B, and it must never be trained with a vanishing compatibility weight, which the code refuses.) For axisymmetric problems there is a simpler system, namely we look for h and λ of the form,

$$h_{ij} = (1+a)\,\delta_{ij} + b\,n_in_j + \tfrac{c}{2}\,(n_iz_j + z_in_j) + d\,z_iz_j, \qquad \lambda = e^u, \qquad (12)$$

with a, b, c, d, u functions of $(t, \mu = n\cdot z)$: 2285 parameters at width 20 and depth 6 with the decay feature (2265 without). The general three-dimensional network of the same size has 2347. Spherically symmetric tests use $h = \alpha(\rho)\delta + \beta(\rho)nn$.

b. Loss. With R_g the residual of group g (Ricci, gauge, scalar equation), the loss is

$$L = \sum_g w_g\,\overline{(\rho^{p_g}R_g)^2} + w_{\rm in}\overline{r_{\rm in}^2} + w_{\rm out}\overline{r_{\rm out}^2} + w_{\rm pin}\sum\langle r\rangle^2 + w_{\rm com}\sum_S |d|^2, \qquad (13)$$

with means over the sampled points. $p = (2,1,2)$ makes every residual dimensionless; for the quadrupolar problem we use $p = (4,3,4)$, an extra weight ρ^4 that determines the far field. $w_{\rm in} = w_{\rm out} = w_{\rm pin} = w_{\rm com} = 100$ (10 for the Weyl tests). The centre-of-mass term uses a fixed product quadrature on each sphere, Gauss–Legendre in $\mu = \cos\theta$ and uniform in ϕ: 24×8 nodes for the axisymmetric network and 48×32 for the three-dimensional one (with data about an axis other than z, the coarser rule aliases the angular structure on the inner sphere; 48×32 agrees with 96×64 there).

c. Sampling points Interior points are drawn log-uniformly in ρ with uniform directions, boundary points uniformly on the inner and outer spheres; all are redrawn every 500 iterations. Every result is checked on five fresh samples, on which the loss agrees with the training loss to within a few per cent (the ratio is 1.00–1.01 for the quadrupolar runs).

d. Training. Adam [23] for 5000 steps, then the self-scaled Broyden quasi-Newton method SSBroyden [24, 25] with a dense inverse Hessian carried between blocks of 250 iterations, a strong-Wolfe line search (Armijo backtracking in the tests of Sec. IV) and scale-invariant curvature guards, in stages of 10000 iterations; a block whose result is not finite is rejected. For the quadrupolar problem one such stage after Adam reaches a loss of

1.1×10^{-10} in 20 minutes. Everything runs in double precision with JAX [26] on NVIDIA A30 GPUs; one value and gradient at 16384 points costs 0.048 s.

e. A spectral reference. For the quadrupolar problem we built an independent approximation: the spherical background (6) plus an axisymmetric, reflection-even perturbation, with Legendre modes $l \le 10$ in the angle and 36 Chebyshev modes in $\ln\rho$ for each of the five metric functions and λ (1008 unknowns), solved by Gauss–Newton on exactly the residual vector of the loss (13). Its loss is 2.8×10^{-11}, its $l=2$ tail decays as $\rho^{-3.00}$ from $\rho=20$ to 100, and evaluated with a network run's own loss on that run's samples it gives 2.8×10^{-11} as well: it is a solution of the same discrete problem, obtained by an unrelated method. Its rotated version grades the off-axis runs.

IV. TESTS WITH EXACT SOLUTIONS

A. The pipeline on exact solutions

Before training anything we check the implementation itself: an exact solution is fed to the loss in place of the network, through the same code (the same derivatives, residuals, boundary terms and weights), and every term must vanish to rounding. For (6) the total loss is below 10^{-16}. For the two-rod Weyl solution it is 1.3×10^{-28} with the gauge source (5); with the harmonic condition ($S=0$) it is not small, since the Weyl chart is not harmonic, which shows that the check is sensitive to the gauge term. The rotated Weyl solution solves the system to 3×10^{-14} or better at every residual. (The averaged pins of Sec. II D are designed for the quadrupolar problem and are not satisfied by these solutions, at the 10^{-9} level; the tests run without them.)

B. Schwarzschild in harmonic coordinates

The spherical network with $R_0=1$, $\lambda_0=0.381966$ on the sphere of h-areal radius 2 and Robin conditions at $\rho_{\rm out}=100$ recovers the exact family. The Robin condition is not set to zero, but rather to the value the exact solutions would produce, so it has a source term. Table I compares the metric-only and first-order formulations on the problem. The metric-only network is more accurate by three orders of magnitude (a factor 2600 in h and 1300 in λ). The first-order network's error is a gauge mode: along it the compatibility residual is nearly flat, and an earlier, less converged first-order run found the same geometry in a different harmonic chart, with h_{rr} off by a third. Deriving Γ from h removes the mode. With the homogeneous Robin conditions (no source), the metric-only network reaches a loss of 2.4×10^{-11} and maximum errors of 7.0×10^{-6} in h, 6.0×10^{-6} in Γ and 6.3×10^{-7} in λ against the exact solution, and its mass $M_0=0.999997$ (Sec. VI) against $R_0=1$; each run takes 9–11 minutes on one GPU. (The centre-of-mass condition plays no role here: a spherically symmetric network has no dipole, and the first-order network's error is a radial chart mode.)

TABLE I. First-order against metric-only network on the same Schwarzschild problem (manufactured Robin source, $\rho_{\rm out}=100$, 20000 Adam + 2000 SSBroyden iterations; `Paper/commands_schwarzschild.sh`).

	first order (h,Γ,λ)	metric only (h,λ)
final loss	1.5×10^{-9}	2.4×10^{-11}
$\max\lvert\delta h\rvert$	1.7×10^{-2}	6.5×10^{-6}
$\max\lvert\delta\Gamma\rvert$	8.6×10^{-3}	7.9×10^{-6}
$\max\lvert\delta\lambda\rvert$	1.3×10^{-3}	1.0×10^{-6}

TABLE II. Weyl rods (masses 1 and 0.1), inner sphere moved towards the rods; 3000 SSBroyden iterations, standard units.

shell	dist. to horizon	$R_{ab}R^{ab}(\rho_{\rm in})$	$\max\lvert\delta h\rvert$	loss
$[7.5,75]$	5.0	3.3×10^{-6}	3.4×10^{-7}	2.4×10^{-14}
$[4,40]$	1.5	5.8×10^{-3}	1.5×10^{-5}	7.1×10^{-13}
$[3.5,35]$	1.0	5.3×10^{-2}	7.3×10^{-5}	3.1×10^{-12}

C. Weyl two-rod solutions

The Weyl class [6] gives exact static axisymmetric solutions with nontrivial multipoles (Sec. II C 2). The inner data, on a coordinate sphere three times the rods' extent for the standard shell, and the Robin sources are taken from the exact solution, and the gauge is (5). For two equal rods (masses 1, gap 1) on the shell $[7.5,75]$ the axisymmetric network trained with SSBroyden from a random initialisation reaches a loss of 2.06×10^{-13} in 3000 iterations, with maximum errors 6.7×10^{-7} in h and 9.0×10^{-8} in λ (Fig. 1). Unequal rods (masses 1 and 0.1) give 2.4×10^{-14}, 3.4×10^{-7} and 5.2×10^{-8}. Their moments are in Sec. VI. Moving the inner sphere towards the rods raises the curvature on it by four orders of magnitude, and the error with it (Table II).

D. A rotated Weyl configuration

Rotating the closest configuration by 45° removes every symmetry. The equations are covariant, so the rotated fields are again a solution, but the gauge is a statement about a chart and the source (5) must be taken about the rotated axis $n=R\hat z$ (Sec. II C 2): with that source the exact rotated solution satisfies the gauge condition to 2×10^{-15}, while it violates the harmonic one ($S=0$) by $|\Gamma^i|\approx2.6\times10^{-2}$. The general three-dimensional metric-only network, with 8192 collocation points and SSBroyden from a random initialisation, reaches a loss of 5.1×10^{-9} in 7000 iterations

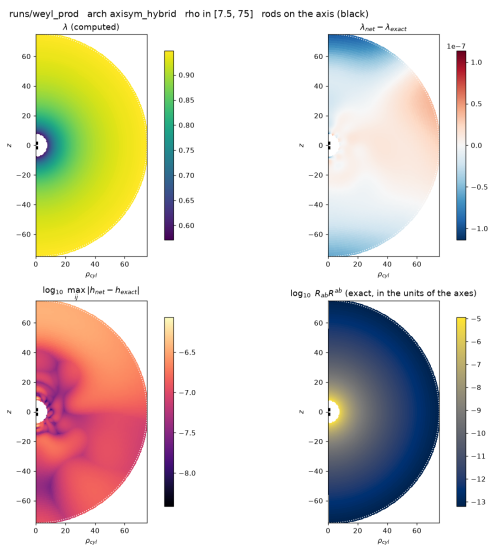


FIG. 1. Equal-mass Weyl rods on the shell $[7.5, 75]$ (meridional plane): computed λ; $\lambda_{\rm net} - \lambda_{\rm exact}$; $\log_{10} \max_{ij} |h_{\rm net} - h_{\rm exact}|$; $\log_{10} R_{ab}R^{ab}$ of the exact solution. [TODO: regenerate without the internal title]

(17 minutes), with maximum errors 1.6×10^{-4} in h and 1.7×10^{-5} in λ, against 7.3×10^{-5} and 1.8×10^{-6} for the axisymmetric network on the same shell.

The general network is about two times less accurate than the axisymmetric one in h and ten times in λ for the same problem; it has to represent a function of three variables instead of two. The number of collocation points matters more for it: the loss is a mean over the sampled points, and with too few of them the optimiser can lower the mean while the residual grows at points it does not sample. We use 8192 points here; an earlier run of the first-order network on 2048 points showed exactly this, its loss falling while every error grew.

V. QUADRUPOLAR DATA

Having tested the solver on exact solutions, we now apply it to a problem without one: spherical data with a quadrupolar perturbation.

A. The problem

We take $R_0 = 1/\sqrt{3}$, the inner sphere $\rho_{\rm in} = 1$ with $r_{\rm in} = 1$ (so $\lambda_0 = 1/3$), $\rho_{\rm out} = 100$, and

$$\lambda|_{\rho_{\rm in}} = \tfrac{1}{3} - \tfrac{1}{12}\, P_2(\cos\theta), \tag{14}$$

a 25% perturbation of the spherical value, with no known closed-form solution. Linear theory predicts a pure $l = 2$ response decaying like ρ^{-3}, no $l = 1$ part and an $l = 0$ back-reaction of order S_2^2. Against the spherical solution the perturbation has an rms size of 2.0×10^{-2} in λ and 2.2×10^{-2} in h near the inner sphere, and 2.6×10^{-4} and 3.5×10^{-6} for $30 < \rho < 100$. We grade against the spectral reference.

B. Without and with the centre-of-mass condition

Two runs with identical settings, except for the centre-of-mass term (Appendix A), show what the condition does (Table III, Fig. 2). Without it, the error is dominated by a part that is odd under $z \to -z$ (an $l = 1$ coefficient of λ peaking at 10^{-5} near $\rho \approx 2$), which the even data forbid; every term of the loss is exactly reflection symmetric, so nothing forces this part: it is the translation mode of Sec. II E. The flux (11) confirms it: it is the same on the spheres $\rho = 1, 2, 10$ and 100 but not zero, $d_z = 7.63 \times 10^{-5}$, and the λ error near the inner sphere is $0.061\,|d_z|$. With $d = 0$ imposed on the spheres $\rho = 1, 1.5, 3, 10, 30, 100$, the error falls by a factor 80 in λ and 70 in h, to 3.1×10^{-8} and 4.2×10^{-7} (rms over the shell), four to five orders below the size of the perturbation. The loss does not see the difference; it is even slightly lower without the condition (5.3×10^{-11} against 9.2×10^{-11}), because the translation costs almost nothing in it. For the same reason its size without the condition is not fixed: it depends on how far the optimisation happens to drift along the mode (in earlier runs with other optimisers it was 300 times larger). The condition removes it.

C. Off-axis data

The same datum about the axis $(1, 0, 1)/\sqrt{2}$ is a fully three-dimensional problem whose solution is the rotated axisymmetric one. With the general metric-only network the error against the rotated reference is [qp_rot: rms λ, h], i.e. [N and M] times below the perturbation. The same network on the unrotated datum gives the same errors after the same budget, so the rotation itself costs nothing; the general ansatz reaches a loss [three to four] orders above the axisymmetric one, the error concentrating in h near the inner sphere.

VI. MULTIPOLE MOMENTS

a. Extraction from the axis. For a static axisymmetric vacuum solution the Geroch–Hansen moments [7, 8, 27] follow from the potential on the symmetry axis in canonical Weyl coordinates, $g = -e^{2U}{\rm d}t^2 + e^{-2U}[e^{2k}({\rm d}W^2 + {\rm d}Z^2) + W^2{\rm d}\phi^2]$: with the static Ernst potential $\xi = -\tanh U$ on the axis, $\xi(0, Z) = \sum_j m_j/|Z|^{j+1}$, the moments are $M_j = m_j$ for $j \le 3$ and $M_4 = m_4 - \frac{1}{7}m_0(m_0 m_2 - m_1^2)$ [28]; equivalently, from $U(0, Z) = \sum_r U^{(r)}/|Z|^{r+1}$, $M_0 = -U^{(0)}$, $M_1 = -U^{(1)}$, $M_2 = \frac{1}{3}(U^{(0)})^3 - U^{(2)}$, $M_3 = (U^{(0)})^2 U^{(1)} - U^{(3)}$ [29].

TABLE III. Error against the spectral reference (angular rms) by radial region, for the same run without and with the centre-of-mass condition (Adam 5000 steps, then 2 × 10000 SSBroyden iterations; the loss is the final training loss).

	loss	λ error				h error			
		1–3	3–10	10–30	30–100	1–3	3–10	10–30	30–100
perturbation size		2.0×10^{-2}	2.5×10^{-3}	7.6×10^{-4}	2.6×10^{-4}	2.2×10^{-2}	4.5×10^{-4}	3.1×10^{-5}	3.5×10^{-6}
without centre of mass	5.3×10^{-11}	4.6×10^{-6}	2.3×10^{-6}	3.5×10^{-7}	4.2×10^{-8}	5.9×10^{-5}	8.6×10^{-6}	8.3×10^{-7}	8.7×10^{-8}
with centre of mass	9.2×10^{-11}	6.4×10^{-8}	6.6×10^{-9}	1.6×10^{-9}	9.1×10^{-10}	8.6×10^{-7}	4.2×10^{-8}	4.7×10^{-9}	3.0×10^{-9}

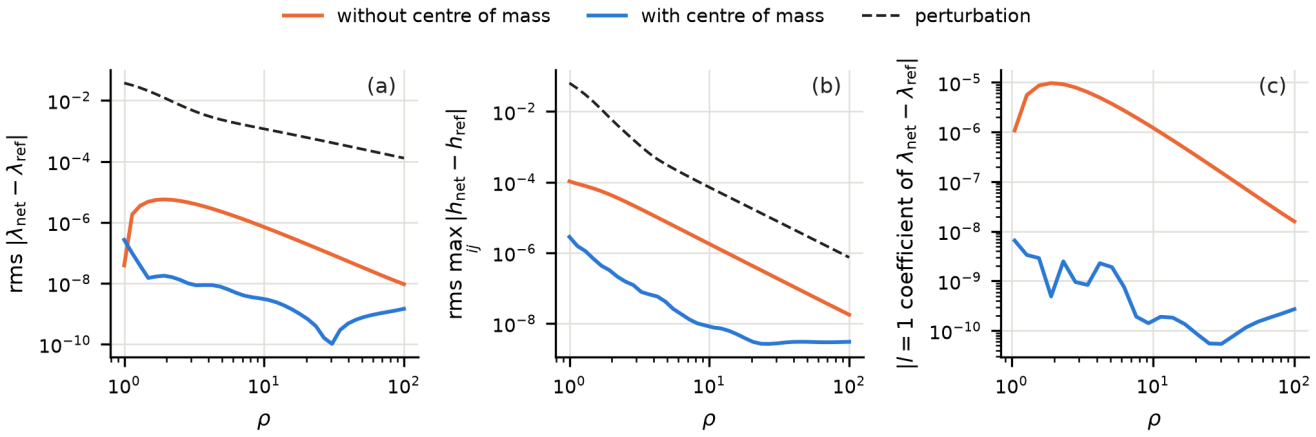


FIG. 2. Error against the spectral reference without and with the centre-of-mass condition: angular rms of $\lambda_{net} - \lambda_{ref}$, of $\max_{ij} |h_{net} - h_{ref}|$, and the $l = 1$ coefficient of $\lambda_{net} - \lambda_{ref}$. Dashed: the perturbation (reference minus spherical solution).

Weyl coordinates need not be built, because only the axis is used: in our variables $U = \frac{1}{2} \ln \lambda$ and $W = |\partial_\phi|_h$; on the part of the axis that reaches infinity $k = 0$, so there $\mathrm{d}Z$ is the proper h-length along the axis; the inner ball cuts the axis in two, and the offset between the branches is the change of Z along a meridian semicircle, $\mathrm{d}Z = \star \mathrm{d}W$ in the meridian metric. We fit ξ on both branches and fix the origin of Z by mass centring ($M_1 = 0$); the error bar is the spread over fit windows starting at 1–1.5 ρ_{in} and fit orders 6 and 8. On exact solutions this returns: Schwarzschild in harmonic coordinates, $M_0 = R_0$ to 10^{-12} and M_l below 10^{-11} for $1 \le l \le 4$; the equal Weyl rods, $M_0 = 2$ and $M_2 = 2.500000$; the unequal rods, the closed-form moments to 10^{-5} relative and the centre of mass $z = 1.309091$; Curzon, $M_2 = -M^3/3$ and $M_4 = 19M^5/105$, although U itself has no quadrupole.

b. The Weyl networks. The trained solutions give $M_0 = 2.000001$ and $M_2 = 2.49950$ for the equal rods (exact 2 and 2.5), and $M_0 = 1.100000$, $M_2 = 0.29105$, $M_3 = -0.5016$ with the centre of mass at $z = 1.3091$ for the unequal rods (exact 1.1, 0.290909, −0.49983, 1.309091). For the rotated rods, extracted about the rotated axis, the three-dimensional network gives (in the chart of the run) $M_0 = 0.0314289$, $M_2 = 6.792 \times 10^{-6}$, $M_3 = -3.26 \times 10^{-7}$ and the centre of mass at 0.03740, against 0.0314286, 6.785×10^{-6}, -3.33×10^{-7} and 0.03740.

c. The quadrupolar solution. Table IV lists the moments. The data raise the mass from the background value $R_0 = 0.57735$ to $M_0 = 0.583847$, a nonlinear back-reaction of 1.1% in the monopole, and the quadrupole moment is $M_2 = 0.1747$. The run without the centre-of-mass condition, whose pointwise error is 80 times larger, has moments as close to the reference as the run with it (within 10^{-6} in M_2): moments are geometric, and the translation mode moves the chart, not the geometry. The extraction shows the mode directly: the centre of mass of that run, measured from the centre of the coordinate sphere of radius ρ, is 1×10^{-7} at $\rho = 1$, 5.8×10^{-5} at $\rho = 2$, 1.17×10^{-4} at 10 and 1.29×10^{-4} at 70: it vanishes at the inner sphere, as the mode must, and tends to the translation $a = d_z/M_0 = 1.31 \times 10^{-4}$. With the condition it is $\sim 10^{-7}$ at every radius.

TABLE IV. Geroch–Hansen moments of the quadrupolar solution: the spectral reference, and the differences network − reference with the same fit windows (the reference's window spread, 1.2×10^{-4} in M_2 and 7×10^{-4} in M_4, largely cancels in the differences). [row for qp_rot].

	M_0	M_2	M_4
reference	0.583847	0.17471	0.0220
with centre of mass	$+1.4 \times 10^{-8}$	$+9.1 \times 10^{-7}$	-7.1×10^{-6}
without centre of mass	-3.5×10^{-8}	$+5.8 \times 10^{-7}$	-2.5×10^{-6}
3-D network, rotated data	[]		

d. Decay of the multipoles. Expanding $\mathcal{P}$ in Legendre modes $c_l(\rho)$ about the solution's own axis, the local power $\mathrm{d}\ln|c_l|/\mathrm{d}\ln\rho$ of the reference is -1.00, -3.00 and -5.0 for $l = 0, 2, 4$ (the last out to $\rho \approx 30$, where $c_4 \sim 10^{-10}$), and its odd modes are at 10^{-16}. The network with the centre-of-mass condition gives the same powers, with its odd modes below 10^{-8} [and qp_rot]. In the mass-centred harmonic chart $c_l\rho^{l+1}$ tends to 0.58387, 0.1747 and 0.0220, i.e. to M_0, M_2 and M_4, consistently with harmonic coordinates being asymptotically Cartesian and mass centred [30, 31]: a second, independent route to the same numbers.

VII. CONCLUSIONS

A PINN solves the exterior boundary-value problem of the static vacuum Einstein equations with prescribed inner data and asymptotic flatness, and reproduces exact solutions with and without symmetry. Two ingredients are essential: a network whose connection is derived from its metric, and a complete gauge fixing. Harmonic coordinates and the inner data leave the asymptotic chart free to translate; the mode is invisible to every residual except at a very low level, and it is the largest error until the pseudo-Newtonian centre of mass is fixed. We expect the same issue, and the same remedy, in any elliptic formulation of the stationary problem whose inner data do not fix the asymptotic frame. The Geroch–Hansen moments of the computed solutions can be read off the symmetry axis without constructing Weyl coordinates, and they are insensitive to that translation. The extension to the stationary equations is the natural next step.

Appendix A: Reproducing the results

[TODO: repository URL and the commit the paper's numbers come from.] The code is the `Stationary` package of that repository: `stationary/train.py` (the solver; every option below is a flag of `python -m stationary.train`), `stationary/specref.py` (the spectral reference), `stationary/gh_moments.py` (the moments). Software: Python 3.14, JAX 0.11.1, Flax 0.12.9, Optax 0.2.8, in double precision (`JAX_ENABLE_X64=1`); hardware: NVIDIA A30 (24 GB). Each result is produced by the commands below, all in the repository.

- Exact-solution checks: `tests/test_pipeline.py`, `verify_weyl.py`, `tests/test_gh_moments.py`.
- Schwarzschild tests (Table I): `Paper/commands_schwarzschild.sh`.
- Weyl tests: the configurations stored with each run (`runs/weyl_*/config.json`); the rotated test, `Paper/commands_weyl.sh`.
- Quadrupolar data: `recipe_pq16k.sh` (the problem) and the stages `qp_com_*`, `qp_nocom_*`, `qp_rot_*`: the Adam stage of `mendieta_submit.sh` (round P; 6 minutes) and the SSBroyden stages of `Paper/hub_qp.sh` (about 20 minutes per 10000 iterations on one A30).
- The spectral reference: `Paper/commands_specref.sh` (reproduces the loss 2.793×10^{-11}; 36 minutes on one A30).
- Grading and moments: `scripts/grade_rotated.py`, `scripts/fresh_sample_loss.py`, `scripts/gh_moments_runs.py`, `scripts/tail_decay.py`.

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